

第三章

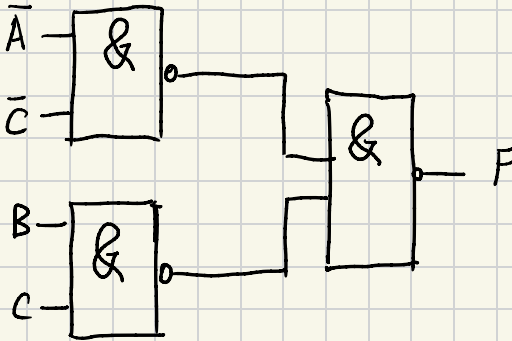
3.1 (1)

AB \ C	00	01	11	10
0	1	1	0	0
1	0	1	1	0

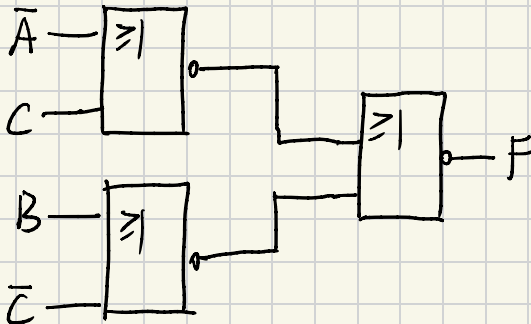
$$\begin{aligned}
 F(A, B, C) &= \bar{A}\bar{C} + BC \\
 &= \overline{\overline{\bar{A}\bar{C} + BC}} \\
 &= \overline{\bar{A}\bar{C}} \cdot \overline{BC}
 \end{aligned}$$

$$\begin{aligned}
 \bar{F}(A, B, C) &= A\bar{C} + \bar{B}C \Rightarrow F = \overline{(A + C)(B + \bar{C})} \\
 &= \overline{(A + C)} + \overline{(B + \bar{C})}
 \end{aligned}$$

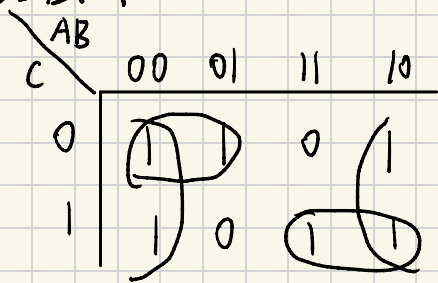
与非门



或非门



(2) 卡诺图为

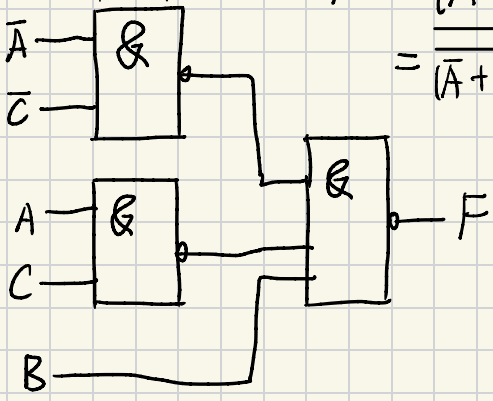


$$F = \bar{B} + \bar{A}\bar{C} + AC$$

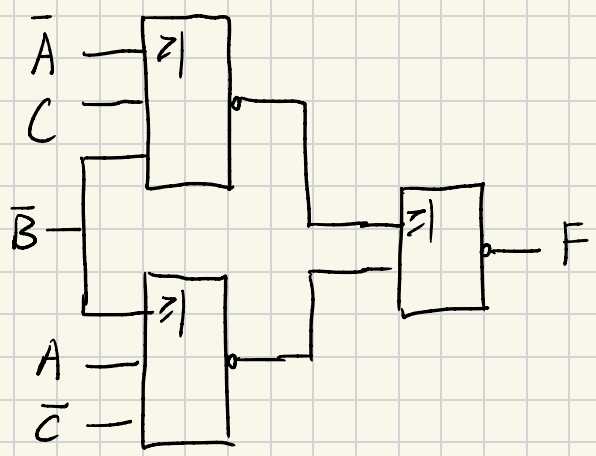
$$= B \cdot \bar{A}\bar{C} \cdot \bar{A}\bar{C}$$

$$\bar{F} = ABC + \bar{A}BC \Rightarrow F = \frac{(\bar{A} + \bar{B} + C)(A + \bar{B} + \bar{C})}{= (\bar{A} + \bar{B} + C) + (A + \bar{B} + \bar{C})}$$

与非门



或非门



(3)

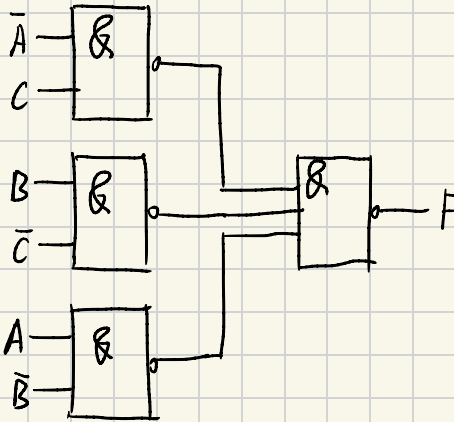
CD \ AB	AB			
	00	01	11	10
00	0	1	1	1
01	0	1	1	1
11	1	1	0	1
10	1	1	0	1

$$F = B\bar{C} + \bar{A}C + A\bar{B} = \overline{\bar{B}\bar{C} \cdot \bar{A}\bar{C} \cdot \bar{A}\bar{B}}$$

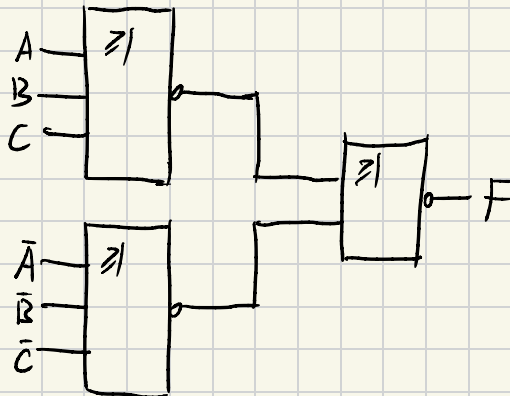
$$\bar{F} = \bar{A}\bar{B}\bar{C} + ABC \Rightarrow F = (A+B+C)(\bar{A}+\bar{B}+\bar{C})$$

$$= \overline{(A+B+C) + (\bar{A}+\bar{B}+\bar{C})}$$

与非门



或非门

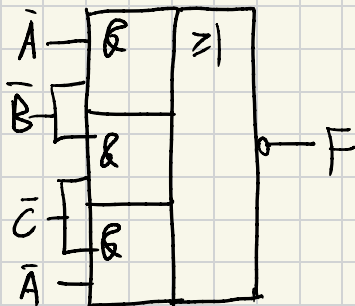


$$\begin{aligned}
 3.2(1) \quad F(A, B, C) &= AB + (A\bar{B} + \bar{A}B)C \\
 &= AB + A\bar{B}C + \bar{A}BC \\
 &= A(B + \bar{B}C) + \bar{A}BC \\
 &= A(B + C) + \bar{A}BC \\
 &= AB + AC + \bar{A}BC \\
 &= AB + C(A + \bar{A}B) \\
 &= AB + AC + BC
 \end{aligned}$$

AB		C			
		00	01	11	10
0	0	0	1	1	0
1	1	0	1	1	1

$$\bar{F} = \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$

$$F = \overline{\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}}$$



3.3. 真值表如下

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

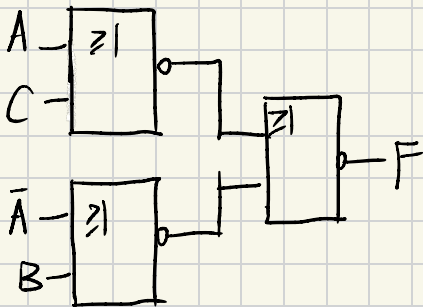
卡诺图如下

AB					
C		00	01	11	10
		0	0	1	0
1		1	1	1	0

$$\bar{F} = \bar{A}\bar{C} + A\bar{B}$$

$$F = (A+C)(\bar{A}+B)$$

$$F = \overline{(\bar{A}+C) + (\bar{A}+B)}$$



3.5 真值表如下

(1)	A	B	C	F	G
	0	0	0	0	0
	0	0	1	1	1
	0	1	0	1	1
	0	1	1	0	1
	1	0	0	1	0
	1	0	1	0	0
	1	1	0	0	0
	1	1	1	1	1

当 $F=0$ 时 ABC 可为 $(0,0,0)$ $(0,1,1)$ $(1,0,1)$ $(1,1,0)$

$F=1$ 时 ABC 可为 $(0,0,1)$ $(0,1,0)$ $(1,0,0)$ $(1,1,1)$

当 $G=0$ 时 ABC 可为 $(0,0,0)$ $(1,0,0)$ $(1,0,1)$ $(1,1,0)$

$G=1$ 时 ABC 可为 $(0,0,1)$ $(0,1,0)$ $(0,1,1)$ $(1,1,1)$

(2)	A	B	C	F	G
	0	0	0	0	0
	0	0	1	1	0
	0	1	0	1	0
	0	1	1	0	1
	1	0	0	1	0
	1	0	1	0	1
	1	1	0	0	1
	1	1	1	1	1

F 功能不变

G 的功能变为当 1 的个数大于等于 2 时 结果为 1

3.7 (1) 由于进行的是平方运算故平方后由原2位数变为四位数, 用 $F_i (i=1, 2, 3, 4)$ 分别表示由高位到低位可以画出真值表如下

A	B	F_1	F_2	F_3	F_4
0	0	0	0	0	0
0	1	0	0	0	1
1	0	0	1	0	0
1	1	1	0	0	1

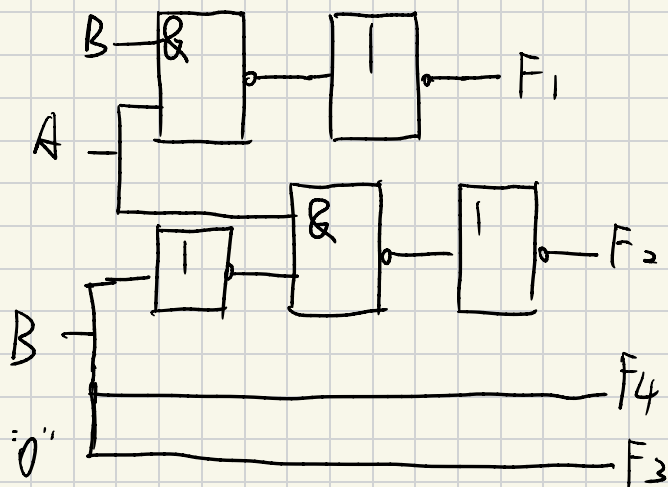
由真值表可分别列出 F_1, F_2, F_3, F_4 的逻辑函数

$$F_1 = AB = \overline{\overline{AB}}$$

$$F_2 = A\bar{B} = \overline{\overline{A\bar{B}}}$$

$$F_3 = 0$$

$$F_4 = AB + A\bar{B} = B$$



3.8. 真值表如下

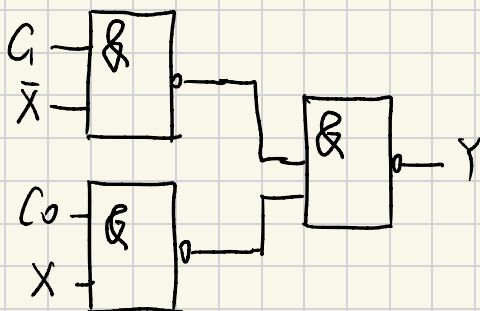
C_1	C_0	Y
0	0	0
0	1	X
1	0	X
1	1	1

根据真值表可以写出 Y 的逻辑表达式

$$Y = C_1 \bar{X} + C_0 X$$

$$= \overline{C_1 \bar{X} \cdot C_0 X}$$

逻辑电路图为



3.9 一位8421BCD码的十进制数, 乘以上之后为8位

设 $X_1 X_2 X_3 X_4 X_5 X_6 X_7 X_8$ 从低位到高位

A	B	C	D	X_8	X_7	X_6	X_5	X_4	X_3	X_2	X_1
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	1	0	1

0	0	1	0	0	0	0	1	0	0	0	0
0	0	1	1	0	0	0	1	0	1	0	1
0	1	0	0	0	0	1	0	0	0	0	0
0	1	0	1	0	0	1	0	0	1	0	1
0	1	1	0	0	0	1	1	0	0	0	0
0	1	1	1	0	0	1	1	0	1	0	1
1	0	0	0	0	1	0	0	0	0	0	0
1	0	0	1	0	1	0	0	1	1	0	1

根据真值表可知 $X_1 = D$ $X_5 = C$
 $X_2 = 0$ $X_6 = B$
 $X_3 = D$ $X_7 = A$
 $X_4 = 0$ $X_8 = 0$

3.10 真值表如下

y_1	y_0	x_1	x_0	z_1	z_2
0	0	0	0	1	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	1
0	1	1	0	0	1
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	1	1

1	0	1	1	0	1
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	1	1

$$Z_1 = \sum_m (0, 4, 5, 8, 9, 10, 12, 13, 14, 15)$$

$y_1 y_0$ $x_1 x_0$	00	01	11	10
00	1	1	1	1
01	0	1	1	1
11	0	0	1	0
10	0	0	1	1

$$Z_1 = \bar{x}_1 \bar{x}_0 + y_1 y_0 + y_0 \bar{x}_1 + y_1 \bar{x}_1 + y_1 \bar{x}_0$$

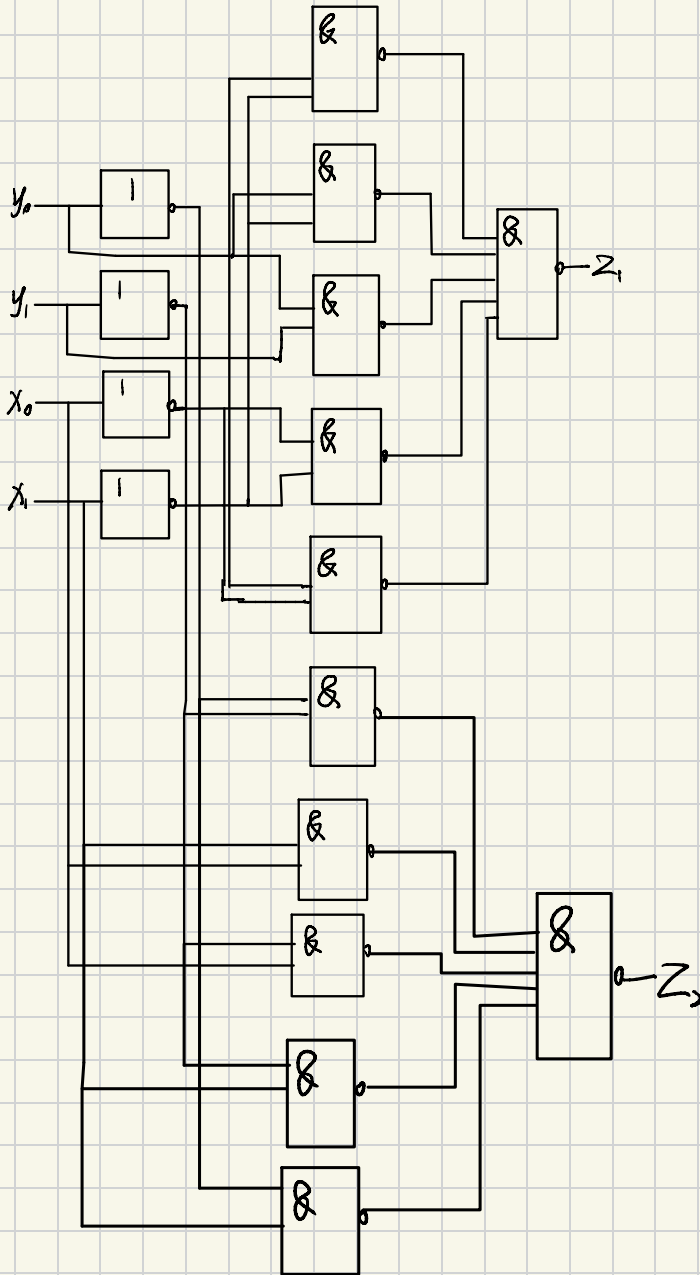
$$= \bar{x}_1 \bar{x}_0 \cdot y_1 y_0 + \bar{y}_1 \bar{x}_1 \cdot \bar{y}_1 \bar{x}_1 \cdot \bar{y}_1 \bar{x}_0$$

$$Z_2 = \sum_m (0, 1, 2, 3, 5, 6, 7, 10, 11, 15)$$

$y_1 y_0$ $x_1 x_0$	00	01	11	10
00	1	0	0	0
01	1	1	0	0
11	1	1	1	1
10	1	1	0	1

$$Z_2 = \bar{y}_1 \bar{y}_0 + x_1 x_0 + \bar{y}_1 x_0 + \bar{y}_1 x_1 + \bar{y}_0 x_1$$

$$= \overline{y_1 y_0} \cdot \overline{x_1 x_0} \cdot \overline{y_1 x_0} \cdot \overline{y_1 x_1} \cdot \overline{y_0 x_1}$$



3.12(2) F_2 的真值表为

AB \ CD	00	01	11	10
00	0	0	1	0
01	1	1	1	0
11	0	1	1	1
10	0	1	0	0

存在冒险 增加冗余项 BD 可消除冒险

$$F_2 = \bar{A}\bar{C}D + ABC\bar{C} + ACD + \bar{A}BC + DD$$

3.13 真值表为

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	1	0	1	1
10	0	1	1	0

$$F = \bar{A}\bar{B}\bar{C} + \bar{A}BD + \bar{B}C\bar{D} + \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{D}$$

$$\bar{F} = (A+B+C)(A+\bar{B}+\bar{D})(B+\bar{C}+D)(A+C+\bar{D})(A+B+D)$$

$$= \overline{(A+B+C)} + \overline{(A+B+D)} + \overline{(B+C+D)} + \overline{(A+C+D)} + \overline{(A+B+D)}$$

